Stress-Testing Product-Gated Clustering and Bounded Priority Ranking for Flood-Rescue Reports

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Abstract. Operational flood streams contain duplicate, incomplete, and adversarial reports, yet clustering and priority scores are often evaluated separately from scheduling consequences. We study a fail-closed batch pipeline: receipt-visible reports with location and event time enter a sparse product-similarity graph; unresolved reports enter review; deterministic duplicate families feed a bounded ranking; and evaluator-only truth joins after predicted-cluster scheduling. A frozen study uses 20 calibration and 40 paired confirmation seeds under in-distribution (ID) and mechanism-shift OOD generators. Independently selected product clustering underperformed additive clustering in ARI by 0.172 ID and 0.100 OOD, and produced 4.11 additional false destinations per 100 reports OOD. The revised priority score was below the legacy score in NDCG@5, while nearest-first scheduling reduced harm and deadline misses relative to the revised policy. Exact duplicate injection caused zero score drift, but coordinated high-confidence nonincident campaigns attained maximal false-priority lift. These negative results delimit what boundedness and geographic edge localization establish: neither implies robust incident consolidation, policy validity, or field benefit.

Keywords: Flood rescue · Graph clustering · Synthetic stress test · Priority ranking · Negative results.

1 Introduction and Related Work

Crisis streams mix repeated observations of one event, nearby but distinct events, missing fields, and nonincident noise. Social-sensor and emergency message research has long emphasized volume, verification, and event extraction [14,3,12]. Multimodal collections such as CrisisMMD and FloodNet capture important disaster modalities [1,13], but they do not jointly provide fine-grained physical-incident linkage, rescue priority, and dispatch outcomes. Consequently, high classification accuracy cannot by itself establish that a report becomes the right operational destination.

Graph community detection [5,16] and density methods such as DBSCAN, HDBSCAN, and ST-DBSCAN [8,6,4] offer plausible consolidation mechanisms. Multiplicative range/domain terms also occur in bilateral filtering [15]; we therefore do not claim a new generic kernel. Spatially constrained clustering and regionalization encode geography through different objectives [7,2]. Humanitarian

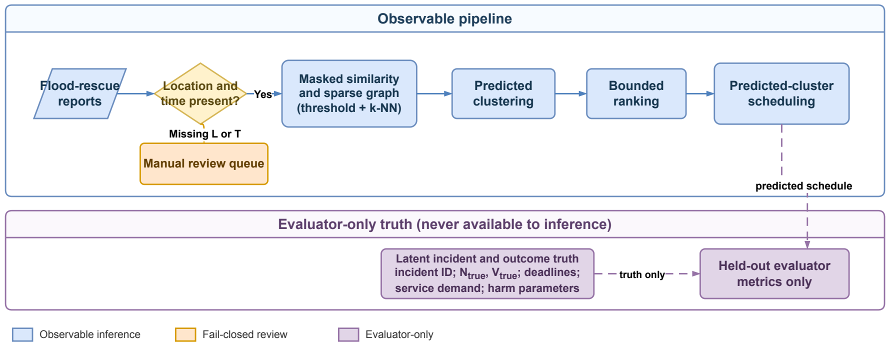

Fig. 1. Observable pipeline (solid), manual-review path (orange), and evaluator-only truth join (dashed). The scheduler receives predicted clusters, never incident truth.

scheduling further exposes conflicting equity, efficacy, and efficiency objectives [9,10,17]. A bounded score is thus a numerical property, not an expert-endorsed rescue policy.

We ask three predeclared questions: (RQ1) does independently selected product composition improve clustering over an additive composition under matched search spaces and one selection rule? (RQ2) does a bounded, duplicate-aware score align with an independently generated benefit and resist missingness, duplicates, contradictions, and campaigns? (RQ3) what happens when predicted split, merge, and noise errors propagate into dispatch rather than being repaired by incident truth?

The contributions are: (i) an observable-only pipeline with explicit review and evaluator boundaries; (ii) a positive-scale product edge-localization statement and a duplicate-aware bounded ranking; and (iii) a frozen paired ID/OOD study that retains adverse results. The main contribution is diagnostic: the confirmation run rejects broad superiority and deployment claims that the same mechanisms could appear to support in an oracle or exploratory study.

2 Observable Pipeline and Methods

Figure 1 fixes the information boundary. Receipt-visible fields flow through graph construction, ranking, and a batch scheduler. Missing location or event time routes a report to review. Incident identity, benefit, deadline, and harm are stored separately and join only after a schedule exists.

2.1 Missingness-Aware Similarity and Graph

A report contains location L_i, event time T_i, flood evidence F_i, urgency evidence E_i, reported demand N_i, vulnerability V_i, provenance features, receipt time, and an observation mask. The graph admission indicator a_i is one only when L_i, T_i

are observed. Missing F_i or E_i is not zero-imputed: let I_F, I_E indicate pairwise shared observations and $m_{ij} = I_F + I_E$; then

$$C_{ij} = \frac{m_{ij}}{2} \exp\left[-I_F \frac{|F_i - F_j|}{\tau_F} - I_E \frac{|E_i - E_j|}{\tau_E}\right], \quad C_{ij} = 0 \text{ if } m_{ij} = 0.$$

For Haversine distance d_{ij} and time difference Δt_{ij}, define

$$\begin{aligned} G_{ij} &= \exp[-d_{ij}^2/(2\sigma^2)], & T_{ij} &= \exp[-\Delta t_{ij}/\tau_t], \\ w_{ij}^{\times} &= a_i a_j G_{ij}(\beta T_{ij} + \gamma C_{ij}), & w_{ij}^{+} &= a_i a_j(\alpha G_{ij} + \beta T_{ij} + \gamma C_{ij}). \end{aligned} \tag{1}$$

All scales $\sigma, \tau_t, \tau_F, \tau_E$ are strictly positive and the weights are nonnegative. The implementation first forms a common spatial candidate pool of at most $\max(64, 4k)$ neighbours per eligible report, including all boundary-distance ties before canonical truncation. It computes an empirical nonzero-weight quantile, retains strict above-threshold edges, keeps the union of endpoint-wise top-k edges, and applies Louvain. Product and additive candidates have the same pool and grid, but each family is selected independently; the estimand is therefore a pipeline comparison, not an operator-only causal effect.

Theorem 1 (Product edge localization). *Let $B = \beta + \gamma > 0$. If a product edge with threshold $0 < \theta < B$ is retained, then $d_{ij} < \sigma\sqrt{2\log(B/\theta)}$. For $\theta \geq B$ the strict retained set is empty; for $\theta \leq 0$ the threshold alone gives no finite cutoff.*

Proof. Since $a_i a_j \leq 1$ and $T_{ij}, C_{ij} \leq 1$, $w_{ij}^{\times} \leq B \exp[-d_{ij}^2/(2\sigma^2)]$. Combining $\theta < w_{ij}^{\times}$ with positivity and taking logarithms gives the bound; the endpoint cases follow from the same inequality.

This is an edge statement. A component-diameter claim additionally needs a bounded observed hop diameter; long transitive chains remain possible. For the additive form, no general threshold-induced geographic cutoff exists when $\theta \leq \beta + \gamma$.

2.2 Duplicate-Aware Bounded Ranking

Provenance $Q_i \in [0, 1]$ combines a source prior, image presence, and capped corroboration from distinct source families; proximity alone cannot increase it. Exact duplicates share a canonical fingerprint. Near duplicates are merged by deterministic complete linkage, so $A \sim B \sim C$ does not merge A with C unless every cross-pair passes the envelope.

Let g index the resulting evidence families. Within each family, observed fields are provenance-gated; across families, demand and vulnerability use maxima rather than unsupported population sums. In compact form,

$$\bar{E} = \frac{\sum_g \max_{i\in g} Q_i E_i}{\max(1, \sum_g \max_{i\in g} Q_i)}, \qquad \bar{F} = \max_{g,i\in g} Q_i F_i,$$
$$\bar{N} = \min\left(1, \frac{\log(1+\max_{g,i\in g} Q_i N_i)}{\log(1+N_{\mathrm{ref}})}\right), \qquad \bar{V} = \max_{g,i\in g} Q_i V_i,$$
$$P = \left[1 + (\mu - 1)\tanh(\bar{V}/s)\right](\omega_E \bar{E} + \omega_F \bar{F} + \omega_N \bar{N}). \tag{2}$$

Claims are clipped before aggregation. We fix $\omega_E = .40$, $\omega_F = .35$, $\omega_N = .25$, and $\mu = 1.75$; hence $0 \leq P \leq 1.75$. The weights and caps are policy knobs, not learned or expert-validated parameters. Comparators are a multiplicity-sensitive legacy score, urgency-only, population-only, a simple linear score, stable random, and—for dispatch—nearest-first.

3 Experimental Design

Data contract and splits. At a predeclared 150-min receipt-time snapshot, reports received by the cutoff are visible; late reports are excluded. A received report with missing event time remains visible but enters review. ID data contain 16 authored incidents; mechanism-shift OOD data vary incident count from 8 to 30 and change report count tails, GPS noise and systematic bias, late order, source/severity-linked missingness, contradictions, duplicates, campaigns, resources, and access delays. Latent benefit is nonlinear, stochastic, and separate from Eq. 2. Seeds are disjoint: 20 development, 20 calibration, and 40 confirmation master seeds, each confirmation seed producing paired ID/OOD datasets. Previously inspected seeds are explicitly retired.

Selection and comparators. Product and additive use the same 128-point nuisance grid: $\sigma \in \{500, 700, 900, 1200\}$ m, $\tau_t \in \{30, 60\}$ min, quantile $\in \{.85, .90, .95, .98\}$, $k \in \{8, 16\}$, and resolution $\in \{.8, 1.2\}$, with fixed $\tau_F = .25, \tau_E = .35$ and $\alpha = \beta = \gamma = .5$. ST-DBSCAN and geo-time HDBSCAN use complete 64- and 96-point grids. All 8,320 calibration executions succeeded. Within each method, one rule maximizes mean ARI, retains the one-standard-error set, then minimizes false destinations, maximizes noise rejection, minimizes review burden, and breaks ties by canonical configuration ID.

Endpoints and inference. Clustering co-primary endpoints are ARI on incident-linked reports [11] and false destinations per 100 reports; noise rejection and review items are secondary. Priority is scored on predicted clusters before a one-to-one evaluator match supplies independent gains; NDCG@5 is primary. Ten stress families cover exact, near, and gradual-chain duplicates; four low-confidence fields; missingness; contradictions; and a coordinated high-confidence nonincident campaign. For dispatch, each predicted product cluster becomes a job; observable centroids, workloads, and three resource scenarios determine the schedule. Truth joins afterward to measure harm, missed deadlines, unreached

Table 1. Clustering descriptives over 40 paired master seeds. FD, NR, and Review denote false destinations per 100 reports, noise rejection, and review items per 100 reports; no adverse or null result is suppressed.

Regime	Method	ARI	FD	NR	Review
ID	Product	0.73	4.98	0.00	16.53
ID	Additive	0.90	4.89	0.00	9.91
ID	ST-DBSCAN	0.85	0.00	0.97	9.66
ID	HDBSCAN	0.85	0.07	0.41	6.95
OOD	Product	0.36	17.13	0.00	45.35
OOD	Additive	0.46	13.02	0.00	37.53
OOD	ST-DBSCAN	0.56	0.41	0.88	43.20
OOD	HDBSCAN	0.56	1.18	0.39	31.18

incidents, false/duplicate trips, tail response, and workload variation. Oracle grouping is a separate diagnostic and never enters inference. Seed-paired effects use 10,000 bootstrap resamples, paired Wilcoxon tests, and Holm adjustment within the frozen clustering and priority/dispatch families. Positive effects always favour the revised candidate; all adverse and null outcomes are retained.

4 Results

Table 1 answers RQ1 negatively. Relative to the independently selected additive pipeline, product ARI was lower by 0.172 (95% CI $[-0.210, -0.136]$) on ID and 0.100 ($[-0.143, -0.058]$) on OOD data; Holm-adjusted $p < 0.001$ in both cases. The ID false-destination difference was inconclusive ($\Delta = -0.10$, $[-0.26, 0.00]$), whereas OOD product clustering emitted 4.11 more false destinations per 100 reports ($\Delta = -4.11$, $[-6.06, -2.35]$, adjusted $p < 0.001$). ST-DBSCAN and HDBSCAN had descriptively higher OOD ARI and much lower false-destination rates than product, but no registered inferential contrast against those baselines supports a superiority statement. Every method lost ARI under mechanism shift.

Table 2, Panel A, also rejects a broad RQ2 alignment claim. Revised NDCG@5 was below legacy by 0.092 ID and 0.068 OOD and was statistically indistinguishable from the simple linear score. It beat population-only and random, but did not consistently beat urgency-only after multiplicity correction. Exact duplicate injection nevertheless caused zero revised-score drift in all 40 ID and 40 OOD datasets. That algebraic success did not authenticate sources: the coordinated high-confidence nonincident campaign produced the maximum false-priority lift, 1.0, in every ID and OOD seed. Thus the duplicate-invariance gate passed, while the general priority-alignment gate remained blocked.

Panel B propagates predicted clustering into RQ3. Against nearest-first, the revised policy's improvement-oriented harm effect was -33.57 ($[-47.88, -19.80]$) ID and -30.88 ($[-53.46, -7.95]$) OOD; deadline effects were likewise adverse, -0.137 and -0.075. Revised ranking reduced harm relative to population-only

Table 2. Paired headline effects with 95% bootstrap confidence intervals. Every Δ is improvement-oriented, so positive favors the revised heuristic; adverse and null effects remain visible. Dispatch values first average the three locked scenarios within each master seed; H is harm and D is deadline-miss percentage points.

Panel A: Priority alignment (revised minus baseline)			
Baseline		ID ΔNDCG@5	OOD ΔNDCG@5
Legacy		-0.09 [-0.12, -0.06]	-0.07 [-0.10, -0.03]
Urgency only		$+0.07$ [$+0.02$, $+0.12$]	-0.04 [-0.09, $+0.01$]
Population only		$+0.44$ [$+0.38$, $+0.49$]	$+0.32$ [$+0.24$, $+0.39$]
Simple linear		-0.01 [-0.03, $+0.01$]	$+0.00$ [-0.02, $+0.02$]
Random		$+0.41$ [$+0.33$, $+0.49$]	$+0.21$ [$+0.13$, $+0.29$]
Panel B: Predicted-cluster dispatch (revised versus policy)			
Policy	Endp.	ID Δ	OOD Δ
Legacy	H	-11.46 [-19.13, -3.20]	-13.65 [-22.59, -4.75]
	D	$+0.74$ [-0.55, $+2.25$]	-1.00 [-2.00, -0.02]
Urgency only	H	$+4.40$ [-4.39, $+14.32$]	$+0.06$ [-13.32, $+15.45$]
	D	-1.83 [-3.60, -0.19]	-0.86 [-1.93, $+0.06$]
Population only	H	$+63.35$ [$+47.88$, $+79.41$]	$+49.53$ [$+18.17$, $+85.96$]
	D	$+3.26$ [$+1.56$, $+5.05$]	$+1.17$ [$+0.26$, $+2.15$]
Simple linear	H	-2.37 [-6.55, $+1.74$]	-2.72 [-9.74, $+5.79$]
	D	-0.47 [-1.33, $+0.39$]	-0.38 [-1.25, $+0.46$]
Random	H	$+59.27$ [$+40.46$, $+79.90$]	$+38.35$ [$+13.39$, $+65.46$]
	D	-0.29 [-2.27, $+1.78$]	-0.19 [-1.56, $+1.13$]
Nearest first	H	-33.57 [-47.88, -19.80]	-30.88 [-53.46, -7.95]
	D	-13.68 [-15.94, -11.51]	-7.54 [-9.66, -5.45]

and random, but not relative to the strongest simple policies. Across the predeclared family and guardrails, the predicted-cluster dispatch-benefit gate therefore remained blocked. These results would have been obscured by oracle incident grouping.

5 Discussion and Threats to Validity

The experiment separates mathematical safeguards from empirical benefit. Product gating localizes retained edges, yet its independently selected pipeline clustered worse than additive and direct density baselines in these authored regimes. Because product and additive selected different nuisance configurations, their contrast is not an isolated causal effect of multiplication. Likewise, score boundedness and exact-duplicate invariance limit numerical influence but do not defend against coordinated independent sources; the campaign result is a concrete counterexample.

All reports, linkage, benefits, resources, and harms are synthetic. OOD changes mechanisms rather than only seeds, but it remains an authored stress family. An audit of planned public crisis and flood sources found no authorized dataset with the joint physical linkage, rescue utility, and dispatch outcomes needed for this evaluation; external, Vietnamese-transfer, real-clustering, and real-dispatch gates remain blocked. No expert panel validated weights or caps. The 150-min batch snapshot omits streaming adaptation, and travel, service, and harm models simplify real access constraints. ST-DBSCAN/HDBSCAN comparisons are descriptive, and the large locked Holm family is conservative. No result supports "deployment-ready", misinformation robustness, or real-world harm reduction.

6 Artifact and Conclusion

The accompanying artifact contains the frozen protocol, environment lock, calibration selection, exact coverage manifests, aggregate analysis, tests, and editable Figure 1. The accepted manifest binds protocol, implementation, selection, raw result, and analysis by SHA-256. The two tables are generated only from the accepted analysis; the oracle artifact and restricted public text are excluded.

In conclusion, the stress test found no product-clustering, general priority, or predicted-dispatch advantage. Product ARI and OOD operational burden were worse than additive; nearest-first outperformed revised dispatch on the main comparisons. Exact-duplicate invariance held, but a coordinated campaign still received maximal false priority. The defensible result is therefore a boundary: edge localization and bounded ranking are useful audit properties, not evidence of incident accuracy, policy validity, or rescue benefit.

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